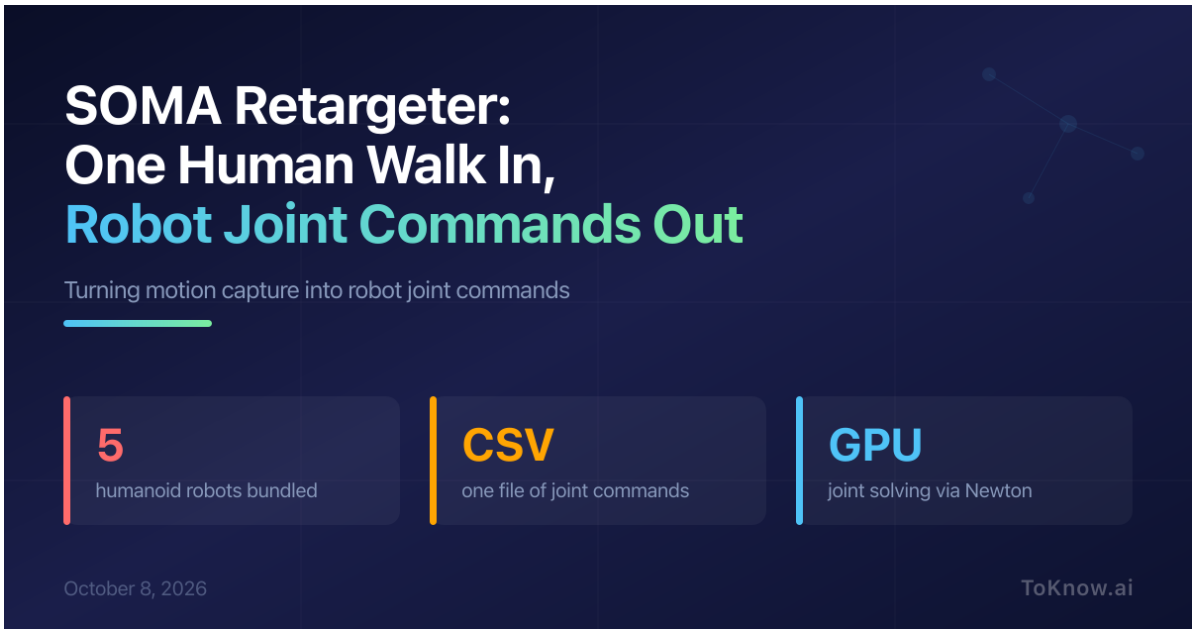
SOMA Retargeter: One Human Walk In, Robot Joint Commands Out

Kabui, Charles

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Read at ToKnow.ai



SOMA Retargeter is an open NVIDIA tool that turns recorded human motion into joint commands a humanoid robot can play back. Give it a motion file built on NVIDIA's SOMA skeleton and it returns a CSV listing the robot's root pose and every joint angle, frame by frame. It ships with five humanoid robots ready to use, among them Unitree's G1 and H2, Booster's T1, and AgiBot's X2 Ultra, plus 10 example walk clips. The September 15 release

added a configurator that registers your own robot from a URDF or MJCF file, plus foot-plant correction and a tool that tunes how the joint angles are solved.

Retargeting is the unglamorous step every humanoid robot project runs into. Motion is captured for people, and a robot has different proportions, joints, and limits, so a recorded walk has to be re-solved as joint angles the robot can follow. Putting that step in one open library, with the solving run on the GPU by NVIDIA's Newton physics engine, removes work every robotics team was doing separately. ROBOTIS already uses it to build motion for its AI Sapiens humanoid. The tool reads motion capture, not video; a single camera enters one step earlier, where NVIDIA's GEM-X estimates full-body motion from one clip.

It is in beta, and the output is joint poses, not a ready controller, so NVIDIA says to validate safety and feasibility in simulation first. The pieces now chain: a body model, a video estimator, and a retargeter take a camera to robot commands.

Read More: Skild AI's S1 turns a video demonstration into robot behaviour

Sources:

- NVIDIA SOMA Retargeter
- ROBOTIS: Soma-retargeter for AI Sapiens
- NVIDIA SOMA body model (SOMA-X)
- Newton physics engine
- NVIDIA GEM-X: whole-body motion from video

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